PHILIP SUH

213-434-1129 | psuh@stanford.edu | linkedin.com/in/phlpsuh | github.com/philipsuh004

EDUCATION

Stanford University Sep 2023 - Jun 2027
B.S. in Computer Science *Stanford, CA*

- **GPA:** 3.89 / 4.00
- **Concepts:** Data Structures & Algorithms, Systems Programming, Machine Learning Fundamentals, Probability Theory

EXPERIENCE

Incoming Software Engineering Intern Jun 2026 - Sep 2026
DoorDash *San Francisco, CA*

- Incoming SWE intern Summer 2026.

Software Engineer May 2024 - Present
Stanford Shen Laboratory *Stanford, CA*

- Designed and evaluated multiple models for automated scanning in microwave impedance microscopy, including particle filter, CNN-based, and multimodal LLM approaches by comparing accuracy on labeled scanning images.
- Improved directional accuracy by 40% and reduced manual scanning time by up to 4 days per sample using a particle filter method which outperformed alternatives.
- Created a GUI that streamlined microscopy workflows and increased throughput for the lab.

Data Science Intern Jun 2024 - Aug 2024
Kavli Institute for Particle Astrophysics and Cosmology *Stanford, CA*

- Analyzed SDSS, Fermi-LAT, and VLBI luminosity datasets in Python (NumPy, SciPy, Astropy) to study accretion disk–jet correlations using non-parametric statistical tests (Kendall's τ, Pearson).

Machine Learning Intern Dec 2023 - Jun 2024
SLAC Atlas Group *SLAC National Laboratory*

- Tested semi-supervised methods for detecting beyond-the-standard-model tracking signatures by evaluating a proprietary permutation-invariant anomaly detection model in Python (scikit-learn), achieving similar $AUC = 0.98$ compared to the commonly used supervised Particle Flow Network.
- Investigated oversampling artifacts in higher-energy physics pile-up simulation data that introduced bias into model training by applying preprocessing techniques using NumPy, Pandas, and SciPy.

PROJECTS

Overwatch Ro.Pa.Sci | XGBoost, TensorFlow, Next.js, TypeScript, PostgreSQL Jun 2025 - Present

- Built a hierarchical esports analytics platform for Overwatch 2.
- Processed time-series game data for fight win rate prediction, first-death classification, and character matchup spreads using an ML pipeline with XGBoost and TensorFlow.
- Optimized PostgreSQL performance enabling ETL pipelines to process 17,000+ data points per game and reducing manual analysis time by 70%.

MIM HyperControl GUI | PyQt5, Sci-kit Image, OpenCV, PyVisa, Matplotlib Jan 2025 - Oct 2025

- Built a modular PyQt5 GUI unifying hardware control into a single interface.
- Generated confidence-weighted scanner movement predictions using a particle-filter method enabling automated navigation across microscopy samples.
- Developed high-performance visualization using `pyqtgraph` for live streaming data and `matplotlib (Qt5Agg)` for embedded scientific plots.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, TypeScript/JavaScript
Frameworks/Libraries: React, Next.js, PyTorch, TensorFlow/Keras, XGBoost, PyQt5, NumPy/SciPy/Pandas/Matplotlib
Technologies: Git, PostgreSQL, Supabase, Multimodal LLM/VLM APIs